ActInf Textbook Group ~ Cohort 1 ~ Meeting 23 (Chapter 10, part 1)

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hello it's active textbook group cohort
one meeting 23 on no 4 22 and we are in
our first of two discussions on chapter
10.
It's the final chapter of the book
and so let's take a look at the table of
contents
and just does anybody from there reading
or experience want to share an initial
note
on
where and how we've come
where do we sit in a satisfaction
satisfaction
space after chapter nine
does chapter 10
get us where we want to be
as we then complete the book
yeah Ali and then I knew what else
actually chapter 10 uh is a pretty
interesting chapter uh with all its
um comparative analysis provided
um and comparing active inference
framework from with other
um similar methodologies and
unsurprisingly emphasizing uh almost
always the advantages of using active
inference over all the other Frameworks
uh but yeah it is comparative analyzes
uh were really helpful in my opinion
in order to distinguish what exactly
active inference is and how it relates
to other uh other disciplinary areas but
uh and I think that some of these
materials could have been introduced

much earlier I mean even in chapters uh
one or two because uh most uh other some
of the confusions at least
some sometimes uh
arises from not being too clear enough
about what actor active inference is and
what it does and how it compares to
other related areas but otherwise it's a
very very interesting chapter in my
opinion
I agree with that yeah I kind of feel
slowly here that the way I would word it
is that
it's almost like a survey uh chapter
but it's got a lot of details and it's
like
that would have been a good maybe not
all of that up front
in the same way but um
it just helps delineate active inference
a little bit more
um
uh just the textbook overall like
I feel like uh
we could make this no you know no
exercises or something and lack of
implementation comment it's like but
uh the fact that it's in
the structure of the way that the you
know
that it is currently understood at the
uh Frontier is almost like a a little
puzzle that will probably result in
significantly better understanding
of the domain and the you know the
so
I feel like it's uh
it's just the right amount of challenge

for me to like really ground what is
being said here and it's not like just
given to you so
um usually that's not done
very well once you have some pedagogical
sort of resource it's like this is what
it is don't think about it just remember
that and that's not very helpful either
so
um
I wish there were maybe other libraries
or something of software that explained
a little more how this was
working in a cogent fashion but
also there's an opportunity to
help write this so that's to me I don't
know more interesting than just uh
wrote definition explanation
yes this isn't the final version
of the textbook
nor the final textbook ecosystem
and I agree yes chapter 10 provides a
nice overview it's it is long and it
also may make sense
as like
chapter 1.1 even
just to understand like the unifying
theory of sentient Behavior
angle
before even talking about how to drive
to active inference
with a high road in the low road
you know
here's the overview in chapter one and
here's Maybe here's how Hawaii is
different than this other
destination
now here are two ways to get there

now here's the generative model and
empirical examples
so many ways to read it Eric any overall
thoughts or let's look at their key
points
um I I feel like this is kind of a
kitchen sink chapter
showing you know trying to
take just a whole conglomeration of
perspectives and lines of of research
and ideas in the uh the whole Enterprise
of computational intelligence
cognitive science neuroscience
some psychology and so forth and say
okay this is how we view it through the
active inference lens
so I think that's the purpose kind of
the purpose of the
the chapters uh the second at least the
second part of the chapter the first
part of the chapter is kind of the
summary of uh the again the overview of
what active inference um is about and
tries to tie the threads together
um
and I don't I don't think it it's it
they're tied any I mean it's good to see
the reminders of you know what what are
the what the journey has been through
the book uh I appreciated that refresh
um there are a number of
you know threads that go in different
directions
um things like you know the notion of
um
everything being a form of inference but
um planning is a different form of
inference because it's a different is a

different functional energy functional
than pure perceptual inference
um
the notion of the exploitation
exploration trade-off
the notion of not being able to solve
for the the base you know optimal
um
policy but um therefore having to use
um a variational
energy approximation that was mentioned
but I feel felt like the book never in
this chapter didn't say well how do we
really leverage that I mean how does
that really come into play
um you know operationally
um
it um reviewed the examples that were
used earlier in the book such as the
team A's and so forth which I found
underwhelming all the way through and
um you know so you know the the chapter
ends with promises it says wow this is
this wonderful Hammer we have let's see
what nails we can find out in the world
and we're going to Hammer Hammer Hammer
and let's see if they behave like nails
you know we're going to treat them like
they behave like nails even if they're
screws
um and you need a different instrument
to turn them
um
so I think that as far as
you know what if if you're a devotee of
active inference then
this chapter is perhaps inspirational
um if you're a skeptic and it's

um it's like well okay okay that's nice
move on
yeah a lot there too let's let's look at
some of these strong claims
one benefit is that it provides of
active inference is that it provides a
complete solution to the Adaptive
problems that sentient organisms have to
solve
of course more could be set but does it
provide a complete solution
or maybe a little bit still generously
but perhaps less excessively
it provides a complete framing or
accounting
which is different than a solution
for the problems
this makes it sound like just you know
install
octave.py and the problem is solved
which is obviously going to be
situational depending on the generative
model which is under no guarantee to be
effective just like no linear model is
effective by principle
so pretty obvious point but something
where as Eric just suggested like a um
adherent
might see this as a slam dunk a fractal
slam dunk
whereas somebody who's approaching this
more skeptically
is actually not going to have seen a
solution at least presented in this book
it offers a unified perspective
on problems like perception action
selection attention emotional regulation
which are usually treated in isolation

in Psych and neuro and addressed using
distinct computational approaches and
artificial intelligence this
much more coherently is true
because it is a unified perspective on
perception learning and but through the
composability of generative models so
merely the particular physics
the Markov blanket itself is not a
unified model of perception action
selection attention and motion
regulation Etc et cetera Etc
however
it's a composable framework under which
these phenomena can be explored
but just the active inference kernel
figure 1.1
is not that but it does offer a
perspective that allows us to integrate
it in a way that linear models aren't
going to get us there and indeed these
fields do work in isolation across
multiple levels of Mars hierarchy
and so a unified perspective is really
important to start having some discourse
about like hey there are two
computational functions from the same
brain region
or other kinds of um
more integrative work so
that's nice
here they're going to contextualize
active inference in light of the
following
other fields Eric first though go ahead
yeah I'll just um add one thing since
you mentioned Mar
um

you know I I actually met David Maher
once um when I was applying for for grad
school and so I was a big part of the
you know I was a big Enthusiast about
the whole school of thought that he he
was the founder of uh at MIT first in in
the UK and then at MIT
and one of the things that some of my
fellow students and I observed was um in
that era was a phenomenon we called
physics Envy which is that um an
observation that physics is the the
grand science uh explained the universe
and it manages to boil things down to
very pithy equations simple equations
that explain a lot you've got
um
E equals MC squared you've got
Schroeder's equation
um you know things like that and people
in computer science and cognitive
science and artificial intelligence
want to have a grand Theory because we
know that the even more interesting
problem for civilization is not how the
universe works it's how do minds work so
we wanted it so at least our professors
and our our mentors we're always looking
for these Grand unified theories little
equations that'll explain everything and
it's important math is important so you
know you can find like equations of um
you know Optical flow and and that it's
really important to model things
um using math but
um it seems that the free energy
principle is is one of these um
physics Envy type of

approaches where we're looking for a
single equation to explain everything
and by the way it does borrow from
statistical physics um so it really is
and it I guess Tristan himself is his
physical background
so from that point of view
um you know this is again
you know looking for a hammer that will
treat the mysteries of nature as a nail
and you can hit it with a hammer and
Presto you've got it and you know it may
be more like biology where there are
lots of principles there's lots of math
to be doing lots of modeling you're
doing but by golly it's really really
complicated and complex and you're never
going to boil it down to
um
you know down the details of how it
actually works
and you know you can you know using any
any simple equations physics does not
predict
um the shape of my coffee cup
I don't care how much you know you know
that that you understand children's
equations or the laws of physics there's
a lot other levels of complexity that
have their own independent principles
and
um you know ways of modeling them that
are required to really understand the
phenomena so my concern is that these
over encompassing claims about active
inference are are falling susceptible to
the the physics Envy
um

track
so sorry that's my oh awesome thank you
yeah well it surely recalls the earlier
um distinction of needs and scruffies
from like chapter one in the book or I
believe
um whether the complexity of any
realized system
is so in the weeds and massively
factorial that
Transcendent generalizations
are ineffectual
um to the physics MV so this is the
classic XKCD
comic much memed
many variants
and um
it also surely represents some
sociological settings like at Santa Fe
Institute and it's like well we can talk
we can have the physicists speculate
about sociology but you wouldn't have
the sociologists speculate about physics
right
and there's cases where that
conversation in both directions is
helpful or not but it just isn't this
projected down number line and then but
as a sort of um rejoinder to the physics
Envy is the fep carrying on this lineage
of physics MV
we may be in a lineage of physics
annexation
because the cognitive physics
is actually expanding to tackle the
scope of classical statistical
thermodynamic and quantum physics
under special cases of cognitive systems

and so I do agree that an individual
might experience physics Envy
or feel like they're getting closer and
closer to the real thing as more and
more equations and and inscrutable
symbols appear
however I wonder if on a deeper
Supra personal level
we're actually aligning physics more
with cognitive science than vice versa
which would completely invert the
traditionally understood
um
physics Envy
they're going to have neuro Envy
or they'll have they'll have mirmika
Envy
Ali
yeah one other case in points in this uh
physics Envy Endeavor is a Karen Broad
uh who's a physicist which tries to
apply her expertise and to uh and to
some of the
um cultural phenomena especially
feminism and she has a very well-known
book uh in that on that subject it's
called meeting the universe halfway and
she's also
the founder of a kind of realism called
agential realism uh which is
a very uh
isomorphic with some of the fundamental
concepts of fep here as well so yeah
there are lots of interminglings between
uh Humanities and Sciences or especially
physics these days
yeah
one image I kind of get is like there's

a micro ratchet towards technicality
applications visibility tangibility
but there's a macro ratchet
towards holism
systems thinking Humanities and so on
and um
without broader awareness of that
situation there can be Whiplash like are
we doing philosophy on what things are
and how they become and becoming an
Unbecoming
or are we calculating a particular form
of a pomdp's expected free energy
functional
and we need to educate
and develop our information environments
to understand both and to include many
perspectives yet on set so 10.2
wrap up just really quickly I'm going to
also say there was an interesting
discussion uh on uh like Lex Friedman's
podcast with Andre carpathiana attention
is all you need the way that they framed
it this is just kind of this
underwhelming kind of mimetic
kind of pithy thing
but they didn't make this claim about
Transformers being kind of some
significantly
a more General uh trainable architecture
that was just kind of in the paper
slipped in and they were just framing it
as here's better uh you know language uh
translation slash prediction right
um
and uh he didn't like that he was like
no it was it was so uh
transformational they should have named

it you know that the Transformers the
new whatever you know and then but then
also had this conversation of
well maybe it worked and it you know
spread because it was this pithy meme
thing that was under promising and then
eventually over delivering I think in
active inference I don't know
I don't see like I see how this like uh
fep
um framing of it is like implicitly over
promising
but I don't see
um a lot of people explicitly constantly
kind of making this like
fep is all you need active full explain
everything sort of framing of it
um but there's this I guess yeah and so
like implicitly
somehow because fep is this sort of
physics framing of it it just it seems
like uh it's
murky because of that relationship but
ultimately we'll just find where it's uh
Well Suited and and not that there
aren't really in fact a lot of males out
there and there may be a lot more screws
or
an equal amount of screws or or we don't
know but we're going to get you know
more complex and more precise models so
how that exactly Works in implementation
is Up For Debate but or it certain means
to be seen But I
I don't know if there's something that
can be done there it's like that's
already modeled as fep as a some sort of
containing background foundational

you know epistemic
um prior there and so they've already
said the you know Transformer is the
amazing thing so it's just it will
deliver
until it meets that or and or not you
know like you know
yep nicely set
like we have the worked example we have
canonical neural networks our active
inference agents we have
the the initial tracers you know the
grappling hooks have been thrown
the bugle has been sounded
now let's see what really happens
so good comments
10.2 wrapping up oh I'll leave please
uh sorry just one point about this claim
in the book that says
the active inference provides a complete
solution to the Adaptive problems uh
well I don't know it uh seems to me uh
like a kind of dubious claim because uh
especially about the word complete here
uh maybe uh it they meant uh that to be
an adequate solution to the Adaptive
problems because uh you see there's a
very uh significant and fundamental
difference between being adequate and
between complete in your Solutions right
so in this case I'm not sure what they
mean by a complete solution to the
adapter problems because uh as we all
know uh even at this stage of of its
development active inference has some
shortcomings and there are some uh well
gaps in the theory and so on and uh so
uh using this strong claim here uh seems

to be uh a little bit out of place and
uh yeah but uh I would be fine if it was
if I have an adequate solution to some
problems in some situations
yeah
10.2 might be a nice reading early
it summarizes
chapters
summarizing chapter chapter one
adaptive exchange the higher and the low
road chapter two the low road Bayesian
inference and generative models that can
be inverted
and heuristics thereof as well as
decompositions associated with those
heuristics
chapter 3.
the high road starting from the
deflationary imperative to preserve
integrity and avoid dissipation
which we may have a more sophisticated
realization of in the Bayesian physics
chapter 5 neurobiology
the most well-studied cases and systems
of interest for many cortical
subcortical motor architectures
supporting the integrative work
pointing to empirical examples on
neurotransmitters composability of
neural systems artificial systems
that's part one of the book chapter six
heads into part two of the book there's
a recipe for Designing systems
seven and eight
are currently summarized as just the
continuous and discrete time settings
and chapter nine talks about model based
data analysis in terms of parametric

recovery of an individual's generative
model in the context of metabasian
inference
connecting the dots 10.3
done it and the whole iguana I guess
usually we would say what the whole
enchilada
but that's not sentient enough
so then it was out there
um
it's a whole cognitive iguana it's a
metabasian iguana we're imagining that
we're building the whole iguana but we
don't no one's actually built the whole
iguana
but also this is so representative how
like threads of research just serve to
be like
like it's like spark after spark with no
fire
like what is it like to be a bat
spawns tens of thousands of directions
from 1954.
but where can you go find what it's like
to be a bat
oh that's right it's still the exact
same speculation that existed in 1954
which is if you had the motor apparatus
and The evolutionary Heritage of being a
bat then you'd know what it was like and
it'd be whatever it is it is like like
but if you're not you're not and so you
don't know
and I don't know if anyone has actually
pierced that Veil
so at least it reflects following up
diligently or more diligently on
provocations from our philosopher

colleagues Eric yeah
I'm sorry this is just a digression you
sent me on well maybe the metaverse will
tell us because then you can have your
virtual reality
environment that gives you only the
abilities the sensory abilities and
motor abilities of a bat and you live in
that for a week and you learn really
what about you know how to think like a
bat what a bad can do can't do
um I don't know just just sorry about
that you no no no no
because that's what I think is the brain
machine interface like where it
literally you'd rewire your brain
to feel it yeah that and then then we'd
know but
you'd have to recover that too
so there's like a difficulty of like
bringing that sensory experience back to
the human brain I don't all right well
obviously yeah might be a no cloning
theorem sort of situation there all
right this is a close enough approach
I'm gonna I'm gonna advertise a piece of
work I did many years ago it's called
the Consciousness machine
and it's um it's a um it's an account
for how to solve a hard problem of
consciousness
the hard problem being how do you uh not
how does a why do we have Consciousness
how does it work what how does it brain
produce it but your own subjective
experience of consciousness
is um the mystery that it's hard to
penetrate through pure science and

explanations so how do you do that you
have to have a user experience you
basically have to make a
it crossed between a video game and a
simulation of your own mind the mind of
your user and
um in that environment then you can not
only see intellectually well how is my
brain working but you can feel it
experience it and make it real to
yourself and in the in the feedback sort
of sense the motor feedback and so once
we have a Consciousness machine which we
don't know how to build yet but we're
getting closer and a matter of uh you
know all the the VR and technology is is
an enabler for that then we'll be able
to
to um you know transform our conscious
experience in various ways including uh
becoming
bats or ants or or whatever by
learning to live those kind of lives in
certain contexts
it's on my website awesome
a short thought on that somebody will
always say well you're not a bat
you're a person thinking you're a bat
but that's where you go to The Meta
Turing test which is now your
in the metaverse can you tell the
difference between the true bat
and the person thinking that they're
about now if you can't design a setting
to differentiate it
from the functional ecological pragmatic
View
it says bat like in the metaverse space

as you can get
so yeah I mean active in the metaverse
and internet of things and and the way
that we can even just today
qualitatively
talk about cyber physical systems
composability intelligence augmentation
distributed cognitive systems these are
all so valuable
and they are not
features that come from
some of the competitors that we're going
to be discussing today
okay the whole iguana
some connections to
uh analyzes of cognition and generalized
challenges in cognitive systems like
exploration exploitation we talked
earlier about how the planning
mediated resolution of explore exploit
was not quite that
but certainly other papers do actually
carry that out now that doesn't mean
that they outperform any other given
model but at least we know that we can
frame explore exploit in a single step
and in multi-step scenarios with active
um and active offers a means of
understanding corresponding neural
computations
the connection between cognitive
processing and expected neurodynamics
was actually very very interestingly
characterized in the recent live stream
number 51.1
which I highly recommend
because this could
be the damn breaking between artificial

neural networks
and active inference pomdp models
so it's quite exciting to hear
we will explore in the rest of this
chapter some implications for
psychological functions as if we were
sketching a psychology textbook
this is kind of like that girdle Escher
Bach style where it's like am I gonna
end the book in the middle of the book
and then fill it with filler I don't
know
has that already happened I don't know
am I sketching a book or is this the
book itself that's up to you
but that's such a funny note
predictive brains Bayesian brain
predictive coating predictive processing
architecture
[Music]
um
these are
popular Frameworks
in
computational cognitive neurosciences
yet also at a deeper level
these predictive approaches
still
struggle for recognition among
somehow default
non-predictive approaches like it always
is needing to be justified in some
people's eyes
whereas
um
active inference takes it several steps
further first by incorporating action
over multiple time skills

although predictive processing via Andy
Clark and others also was making
directions in that way but in a less
quantitative way
and also introduces this variational
physics grounding
like
4E cognition plus predictive processing
with inference in action
simply smashing those two together in
your theory Collider
would never reveal that classical
statistical thermodynamic and quantum
mechanics can be understood as special
cases of a cognitive physics
so that's sort of this like third vector
that is exceptionally technical but does
more than or makes active inference more
than merely for ecognition plus
predictive processing as qualitative
syntheses with
attractable implementation approach in
the pomdp
for example
perception and learning
we've talked about how there are similar
inferential processes playing out over
different time scales and that maps to
some biologically plausible
physiological updates in terms of
synaptic plasticity for Learning and
neural firing rates for perception
and it's a very strong mapping indeed
because the imperative in terms of the
loss function of the neural network
being trained
includes the firing and the plasticity
term

and that actually brings a very strong
synthesis between the loss function in
the neural networks training and the
variational free energy descents in the
pomdp
so
figure one
in isomera at all
and the loss function on a neural
network as a function of observations
and parameterization
is concordance with the variational free
energy in terms of observations and
parameterizations
and
we can model
one interesting piece here I know this
is not the textbook itself but this
fictive casuality causality
is like the opposite of the expected
free energy
so it's almost like variational free
energy is the now casting
expected free energy is anticipated
decision making
and fictive causality is looking only at
the past
and it has a computational
implementation in terms of a risk
variable
and a neuromodulatory interpretation
with respect to dopamine modulation
so very interesting and um tractable
Bayesian brain
I don't know what more can be said it's
about time that the Bayesian brain got
formal
but there's a lot of Pros on the

Bayesian brain
and a lot of
reprinting of the Bayes equation
but not much in the way of meaningful
contributions
beyond that
and where there are meaningful
contributions they're not carried out in
the spirit of the unified sentient
behavioral model it's just like oh this
brain region can be understood as doing
prior updating on this inputs
so it's using Bayesian framework
Bayesian statistics to describe
neurophysiology but that is
unless superintentionally otherwise
on the scruffy end of the spectrum
whereas active is like coming from the
neat end of the spectrum and connecting
some of those scruffy findings
um action
both action processing and perceptual
processing are Guided by forward
predictions
many contact points with various motor
embodied embedded perspectives that
we've discussed ecological psychology
affordances cups are for grasping change
the world change your mind
we're talking about action
the consequences of past action what to
do now
planning in the future
idea motor theory that's an interesting
thread with William James
and um Beyond
and it is about connecting ideas with
behavior

that is getting formalized here
cybernetics
um the tote as well as the UDA Loop
observe Orient decide Act and the test
operate test exit model
and cybernetics brings in the concept of
the goal-oriented uh telios
and in certain cybernetic settings that
was operationalized further as
perceptual control theory which is that
what is being bounded
is surprise about perception
interestingly and as um
Dr isamara described in our conversation
the Bayesian risk is calculated about
uncertainty on hidden States
which allows the Bayesian risk to be fit
monotonically and with a um heuristic
approximation
and in doing so prevents overfitting of
observables
so in the fully observable chessboard
situation of course observations in
Hidden states are one of the same and
the a matrix is just an identity Matrix
mixing our metaphors
whereas when there's uncertainty between
the hidden State and the outcome
one may be down a fallacious cul-de-sac
if they overfit on the observations
because there might be like Amanda
mapping or many many mapping or
stochastic mapping between hidden States
and observations however
under an appropriately structured
partition
the particular partition
in particular

one may fit a central tendency and
variance attribute on an unobserved
state
and by doing so resists overfitting
categorically to outcomes
and also enables fitting of large or
sparse data sets of observables
optimal control theory
discussed in various previous live
streams and cases
um
how our inverse and forward
control theoretic
in the control theory ontology
how does it map to the way that we talk
about
um message passing and active inference
General key point the cost or value
function
is um
a special case or part of the
decomposition of a variational or
expected free energy
however we're not shoehorning everything
into cost or value
however again components or special
scenarios of the free energy can be
understood to have a pragmatic value
like or a cost like
components
utility and decision making
dynamic programming and the Bellman work
some of these ad hoc
bonuses novelty bonus intrinsic reward
and so on
are being approached in a first
principle's way through
unified imperatives like the expected

and variational free energy which can be
decomposed into pragmatic and epistemic
components
so whence utility oh utility where art
thy sting how am I going to get utility
by learning about active inference
Bayesian decision making coming closer
from the cybernetic and utilitarian
decision-making space into specifically
the Bayesian brain and Bayesian
decision-making space this Bears very
very heavily on the complete class
theorem and so Ali thanks a lot for um
raising this
in the um preparations for 51.
because the Bayesian beliefs can be
framed in terms of willingness to bet
willingness to pay
but importantly from the complete class
theorem perspective as basically like an
actionability
and the specific set of strategies
that they are interested in and restrict
their analysis to are the admissible
decision rules
which means that that decision rule is
as good as other decision rules
everywhere or at least somewhere
so by staying in subspaces of admissible
rules
one has strategies that are in some way
functional
meaning more functional than
alternatives
which is very interesting because that
is still a large space with many bizarre
strategies some of which are very
fragile

but their admissibility
is a constraint
that is equivalent to saying that the
calculation of perception is going to be
Bayes optimal
like there's some signal input
for which the common filter of time
Horizon 3 7 11 is going to be the the
um Bayes optimal solution or for
whatever length of Kalman filter or
Bayesian filtering approach you're
taking you can calculate it Bayes
optimally
and that's in the inference space and
then analogously we can think about the
admissible strategies in the strategy
space
some reinforcement learning
addressed but subtle
reinforcement and reward
are
decompositions of expected and
variational free Energies
and um
so
it's not that we don't reinforce
high value behaviors it's just that the
model architecture is quite distinct
they then go into these families of RL
model free reinforcement learning model
based and policy gradients
some similarities and differences with
all of these
planning is inference which we may or
may not have seen in the book depending
on how you think about it
but the key point is that as future
outcomes haven't happened

even hypothetically
one has to take a different approach
towards planning
and the consequences of hypothesized
consequences of potential actions or
policies
it's a slightly different topic or
question than inferring
the relationship between actions taken
in the world and the consequences on how
things change
and it actually is captured very
interestingly again
with the um
fictive causality
which can be understood as like a
retrospective learning
from just starting with the first data
point already you're in the game
but starting from even the first round
and then accumulating through training
learning the relationship of actions
and
transitions in the world and allowing
re-evaluation thereof
because some actions have delayed
consequences and so on
learning that in an open-ended way
while actions truly do have consequence
in the real world
in the generative process
so active inference takes a unique
approach for planning
first we use variational inference
second it's model based planning third
we use an integrated expected free
energy functional
that subsumes other schemes like KL

optimal control
so out at worst
it should fit these other schemes as
well as they do
albeit with some computational overhead
at best we should strictly be better
behavior and bounded rationality
Cicero
bounded rationality and free energy
interestingly free energy Theory
more of the psychology valence emotion
motivation
Dante
homeostasis allostasis enteroceptive
processing
we just completed 50
1.2 yesterday interception as modeling
allostasis as control
this is an important area
and these models are fit
attention salience and epistemic
Dynamics so we talked we saw a kind of
like utility oriented above with the
bounded rationality and the
reinforcement learning and so on and now
we're moving towards the epistemic
components being addressed in a first
principle's way
um salience and attention
now into the structural
learning
rule learning causal inference fast
generalization
building on the Bayes graph causal
architectures and thinking about
inference and inferring fast and slow
meta learning
active inference and other fields open

directions
active inference can be applied to other
domains two domains social cultural
Dynamics and
ML and robots
social and cultural thinking through
other Minds
the anticipating brain is not a
scientist ecological cognitive processes
Birdsong
which as we've discussed isn't also
the communication example that people
think it is it's an example of
diatically embodied
pre-scripted
monologue
that's why it's called a duet for one
this is not open-ended
symbolic message passing
so much much more needs to come into
play
before we can say that we have active
inference models of like improvising
entities but some of the turn taking
architecture is is adequate and for
example these birds expect to hear a
certain song but we could use this model
to say well we just expect there to be a
conversation
we expect little dead air
so we can use this
mechanics
to simulate the Dynamics of turn taking
but we need a different payload for the
actual communication if we want those
entities those birds to be doing
anything other than just
saying what they already expect each

other to say
and in the machine and cumulative
culture and all of these complexities
extended cognition
machine learning and Robotics
JF is the one to speak to on this topic
and it's really interesting with the
cognitive robotics angle
how like
it's helping get action fully into the
loop
and by doing so showing which aspects of
the inference and action are embodied
embedded and cultured Etc et cetera Etc
and we end
with the Lord of the Rings
quote
perhaps referencing the one ring on the
cover of the textbook perhaps not
they started
by daring to be naive and ask whether it
is possible to understand brain and
behavior from first principles
and brought active inference from off
stage
to address that question
they
hope
expect and prefer
that the reader has been convinced that
the answer to the original question is
yes it is possible to understand brain
and behavior from first principles
the book is a useful compliment
to philosophy and physics Kristin
a physics for a particular systems
2019 monograph
currently under contract at the same MIT

press
and speculatively we can say that this
will be the active inference textbook
and there is forthcoming a physics
textbook
that will be more about the fep
the journey ends
it was an introduction
you can't just learn it in theory
in theoretical neurobiology
you have to get your hands on the models
and their misbehaviors
so like the classic Turing
quip
when someone said how can computers be
creative when they're only put out what
we put in they never surprise us we're
just getting you know we're just getting
what we put in we hear that today
entering said I'm surprised by computers
every day
whether or not you're going to be
modeling computationally
you may reflect on it day to day
it might be about where you look
it might be about where you lick
it might be about what you do
ultimately we are confident that you
will continue to pursue active inference
in some form
Bravo to those who
stay through the cohort
do you have any last thoughts for today
yeah well excellent reading thank you
thank you for going through it in such
detail that's a great summary of your
part
I'm trying to I'm trying to you know

balance the um the epistemic and the
pragmatic which is to say the comedic
value
um
we'll talk again about chapter 10 maybe
go into a few more details and then
probably highlight
projects
and the ways that we can return in 2023
for cohort three
with something like a cleaner slate
as we welcome the next set of people in
we accreted a lot we got all the
equations in we got all the figures in
all the boxes are in
now we have the the actual task laid out
in terms of
space
so we'll be ready to proceed
all right
going to finish the recording now thank
you guys